Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 33: Approved Solutions

The Pigeonhole Principle

Covers Chartrand & Zhang, Section 8.3

Status of this set

Ungraded practice. Solutions released with it. For every problem, **name the pigeons and the holes explicitly** — that is the graded part.

Part I: How to Practice This Material

Three things to identify

1. The **pigeons** — the objects being placed.
2. The **holes** — the categories.
3. The **rule** assigning each pigeon to a hole.

Then: n pigeons in k holes forces some hole to hold at least $\lceil n/k \rceil$. To guarantee r in some hole you need $k(r-1)+1$ pigeons.

Choosing the holes is the creative step

The holes are rarely handed to you. Group the possible values so that any two items landing in the same group have the property you want — consecutive, summing to a target, sharing a remainder.

Three mistakes to expect

1. **Not naming the holes.** "By pigeonhole, two must match" is not an argument.
2. **Off-by-one on $k(r-1)+1$.** With $k(r-1)$ items the conclusion can fail.
3. **Claiming to know *which* two.** Pigeonhole proves existence, not identity.

Part II: Practice Problems

Section A — Basic counts

1. Answer each, naming the pigeons and holes.

(a) A drawer holds socks of 3 colours. How many must you draw to guarantee a matching pair?

(b) In a class of 30, how many must share a birth month?

(c) How many cards must you draw from a standard deck to guarantee 4 of the same suit?

Approved solution

(a) Four. Pigeons: socks drawn. Holes: the 3 colours. With 3 socks you could have one of each; a fourth forces a repeat. By the formula, $k(r-1)+1 = 3(1)+1 = 4$.
(b) At least three. Pigeons: 30 students. Holes: 12 months. $\lceil 30/12 \rceil = 3$.
(c) Thirteen. Pigeons: cards. Holes: 4 suits. $k(r-1)+1 = 4(3)+1 = 13$. With 12 you could have exactly three of each suit.

In each part the second sentence — what could go wrong with one fewer — is what shows the bound is sharp.

2. Show that among any 8 integers, two of them differ by a multiple of 7.

Approved solution

Pigeons: the 8 integers. **Holes:** the 7 possible remainders on division by 7, namely $0, 1, \ldots, 6$. **Rule:** send each integer to its remainder.
Since $8 > 7$, two integers a and b land in the same hole, so they have the same remainder. Then $a - b$ is divisible by 7. ■

Remainders are the most useful holes in this subject. Whenever a problem mentions divisibility or differences, try them first.

Section B — Choosing the holes

3. Show that if 6 integers are chosen from $\{1, 2, \ldots, 10\}$, then two of them sum to 11.

Approved solution

Holes: the five pairs

$$\{1, 10\},\ \{2, 9\},\ \{3, 8\},\ \{4, 7\},\ \{5, 6\},$$

each of which sums to 11. Every integer from 1 to 10 lies in exactly one pair, so these are 5 pairwise disjoint holes.
Pigeons: the 6 chosen integers. Since $6 > 5$, two land in the same pair — and the two members of a pair sum to 11. ■
Sharpness: with 5 integers it can fail. Choosing $\{1, 2, 3, 4, 5\}$ takes one from each pair, and no two sum to 11.

4. Show that if 7 integers are chosen from $\{1, 2, \ldots, 12\}$, then two of them sum to 13. Then show 6 is not enough.

Approved solution

Holes: the six pairs $\{1, 12\}, \{2, 11\}, \{3, 10\}, \{4, 9\}, \{5, 8\}, \{6, 7\}$, each summing to 13. **Pigeons:** the 7 chosen integers. Since $7 > 6$, two share a pair and sum to 13. ■

Six is not enough: $\{1, 2, 3, 4, 5, 6\}$ takes one from each pair, and no two of these sum to 13.

The same construction, with the pairing changed to match the target sum. Recognising the family a problem belongs to is most of the work.

Section C — Writing

5. In five to eight complete sentences, explain why the pigeonhole principle is a *non-constructive* method. Use one of the problems above, and connect your answer to Lecture 26.

Approved solution

Model answer. A pigeonhole argument establishes that two objects must share a hole by comparing counts, not by examining the objects themselves. In Problem 3, for instance, we know that some two of the six chosen integers sum to 11 because six items cannot fit one-to-one into five pairs — but the argument never identifies which two, and it cannot, since the answer depends on which integers were chosen. That makes it a non-constructive existence proof in the sense of Lecture 26: it proves an object exists without producing it. The contrast with a constructive proof is sharp, because a constructive argument would amount to a procedure for finding the pair, and pigeonhole supplies none. For a computer scientist the distinction matters, because knowing a collision exists is not the same as being able to locate it efficiently.

Full credit requires naming the specific problem and saying why the argument cannot identify the pair.

Section D — Challenge (optional)

6. Show that in any group of $n \geq 2$ people, two people have the same number of acquaintances within the group. (Assume acquaintance is mutual.)

Approved solution

Pigeons: the n people. **Holes:** the possible acquaintance counts, which range over $0, 1, \ldots, n-1$ — that is n holes, so the principle does not apply yet.
The observation that fixes it: the counts 0 and $n-1$ cannot both occur. If someone knows nobody, then nobody knows everybody; if someone knows everybody, then nobody knows nobody. So at most $n-1$ of the n counts are actually available.
Now there are n people and at most $n-1$ possible counts, so two people share a count. ■

This is the classic pigeonhole problem, and the difficulty is entirely in noticing that two holes are mutually exclusive. The count of holes is not always what it first appears.

Instructor notes

- **Problems 3 and 4 are a matched pair** — same construction, different target sum. Assign both; the second should take a minute once the first is understood.
- **Problem 6 is the best item here.** Two holes cannot both be occupied, so n holes become $n-1$. Worth board time; students find it genuinely surprising.
- **Problem 5** connects pigeonhole to Lecture 26's non-constructive existence.

- **Insist on naming pigeons and holes** throughout. An answer without them is not a proof, and Homework 12 grades on it.
- Reserved for Homework 12: Section 8.3 even-numbered exercises.